

memory items + systems telemetry

budget knob (L, C, B)

Perception: GNN over memory graph (co-access / entity edges)

Factored policy $\pi_{rep} \cdot \pi_{tier} \cdot \pi_{time}$

π_{rep}

π_{tier}

π_{time}

masked softmax over 50 legal actions

Online execution
MemCube migrate/fuse
LMCache put/get

Offline (sleep-time)
consolidation queue:
plaintext $\rightarrow$ latent/param

Substrates: MemOS store | LMCache tiers (GPU/CPU/disk/remote) | M+ latent

Critic
(value head)
under (L,C,B)

reward $r = U(\text{task utility}) - \lambda_L \text{lat} - \lambda_C \text{cost} - \lambda_B \text{tok}$

credit assignment